Current State	opcode [6:0]	func3 [2:0]	func7[5]	alu_zero	mem_ready	rf_chip_en	rf_write_en_n	mem_addr_src	mem_write_unit_	mem_read_en	ir_write_en	pc_write_en	rf_write_src [1:0]	alu_b_src	alu_opcode [3:0]	pc_src [1:0]	Next State
Fetch	x	x	x	x	0	1	1	0	0	1	0	0	0	0	0	0	Fetch
Fetch	x	x	x	x	1	1	1	0	0	1	1	0	0	0	0	0	Execute1
Execute1	0110011 (R)	0	0	x	x	1	0	0	0	0	0	1	0	0	0000 (ADD)	0	Fetch
Execute1	0110011 (R)	0	1	x	x	1	0	0	0	0	0	1	0	0	0001 (SUB)	0	Fetch
Execute1	0110011 (R)	1	x	x	x	1	0	0	0	0	0	1	0	0	0010 (SLL)	0	Fetch
Execute1	0110011 (R)	10	x	x	x	1	0	0	0	0	0	1	0	0	0100 (SLT)	0	Fetch
Execute1	0110011 (R)	11	x	x	x	1	0	0	0	0	0	1	0	0	0110 (SLTU)	0	Fetch
Execute1	0110011 (R)	100	x	x	x	1	0	0	0	0	0	1	0	0	1000 (XOR)	0	Fetch
Execute1	0110011 (R)	101	0	x	x	1	0	0	0	0	0	1	0	0	1010 (SRL)	0	Fetch
Execute1	0110011 (R)	101	1	x	x	1	0	0	0	0	0	1	0	0	1011 (SRA)	0	Fetch
Execute1	0110011 (R)	110	x	x	x	1	0	0	0	0	0	1	0	0	1100 (OR)	0	Fetch
Execute1	0110011 (R)	111	x	x	x	1	0	0	0	0	0	1	0	0	1110 (AND)	0	Fetch
Execute1	0010011 (I)	0	x	x	x	1	0	0	0	0	0	1	0	1	0000 (ADD)	0	Fetch
Execute1	0010011 (I)	1	x	x	x	1	0	0	0	0	0	1	0	1	0010 (SLL)	0	Fetch
Execute1	0010011 (I)	10	x	x	x	1	0	0	0	0	0	1	0	1	0100 (SLT)	0	Fetch
Execute1	0010011 (I)	11	x	x	x	1	0	0	0	0	0	1	0	1	0110 (SLTU)	0	Fetch
Execute1	0010011 (I)	100	x	x	x	1	0	0	0	0	0	1	0	1	1000 (XOR)	0	Fetch
Execute1	0010011 (I)	101	0	x	x	1	0	0	0	0	0	1	0	1	1010 (SRL)	0	Fetch
Execute1	0010011 (I)	101	1	x	x	1	0	0	0	0	0	1	0	1	1011 (SRA)	0	Fetch
Execute1	0010011 (I)	110	x	x	x	1	0	0	0	0	0	1	0	1	1100 (OR)	0	Fetch
Execute1	0010011 (I)	111	x	x	x	1	0	0	0	0	0	1	0	1	1110 (AND)	0	Fetch
Execute1	0110111 (LUI)	x	x	x	x	1	0	0	0	0	0	1	0	1	1111 (PASS_B)	0	Fetch
Execute1	0010111 (AUIPC)	x	x	x	x	1	0	0	0	0	0	1	11	0	0	0	Fetch
Execute1	0000011 (LOAD)	x	x	x	x	1	1	0	0	0	0	0	1	1	0000 (ADD)	0	Execute2
Execute1	0100011 (STORE)	x	x	x	x	1	1	0	0	0	0	0	0	1	0000 (ADD)	0	Execute2
Execute1	1100011 (BEQ)	0	x	1	x	1	1	0	0	0	0	1	0	0	0001 (SUB)	1	Fetch
Execute1	1100011 (BEQ)	0	x	0	x	1	1	0	0	0	0	1	0	0	0001 (SUB)	0	Fetch
Execute1	1100011 (BNE)	1	x	0	x	1	1	0	0	0	0	1	0	0	0001 (SUB)	1	Fetch
Execute1	1100011 (BNE)	1	x	1	x	1	1	0	0	0	0	1	0	0	0001 (SUB)	0	Fetch
Execute1	1100011 (BLT)	100	x	0	x	1	1	0	0	0	0	1	0	0	0100 (SLT)	1	Fetch
Execute1	1100011 (BLT)	100	x	1	x	1	1	0	0	0	0	1	0	0	0100 (SLT)	0	Fetch
Execute1	1100011 (BGE)	101	x	1	x	1	1	0	0	0	0	1	0	0	0100 (SLT)	1	Fetch
Execute1	1100011 (BGE)	101	x	0	x	1	1	0	0	0	0	1	0	0	0100 (SLT)	0	Fetch
Execute1	1100011 (BLTU)	110	x	0	x	1	1	0	0	0	0	1	0	0	0110 (SLTU)	1	Fetch
Execute1	1100011 (BLTU)	110	x	1	x	1	1	0	0	0	0	1	0	0	0110 (SLTU)	0	Fetch
Execute1	1100011 (BGEU)	111	x	1	x	1	1	0	0	0	0	1	0	0	0110 (SLTU)	1	Fetch
Execute1	1100011 (BGEU)	111	x	0	x	1	1	0	0	0	0	1	0	0	0110 (SLTU)	0	Fetch
Execute1	1101111 (JAL)	x	x	x	x	1	0	0	0	0	0	1	10	0	0	1	Fetch
Execute1	1100111 (JALR)	0	x	x	x	1	0	0	0	0	0	1	10	1	0000 (ADD)	10	Fetch
Execute2	0000011 (LOAD)	x	x	x	0	1	1	1	0	1	0	0	1	1	0000 (ADD)	0	Execute2
Execute2	0000011 (LOAD)	x	x	x	1	1	0	1	0	1	0	1	1	1	0000 (ADD)	0	Fetch
Execute2	0100011 (STORE)	x	x	x	0	1	1	1	1	0	0	0	0	1	0000 (ADD)	0	Execute2
Execute2	0100011 (STORE)	x	x	x	1	1	1	1	1	0	0	1	0	1	0000 (ADD)	0	Fetch